Lisul Elvitigala

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Education

Stony Brook University Expected Graduation: May 2028

B.S. Computer Science and Applied Math & Statistics, Presidential Scholar *GPA: 3.86/4.00*

- Coursework: Algorithms, Data Structures, Probability & Statistics, Linear Algebra, Differential Equations
- **Honors:** MIT Micromouse Judge (1 of 4 selected); 2x FIRST Robotics World Championship Qualifier

Experience

Hitachi Rail June 2026 – Aug 2026

Software Engineer Intern, Onboard Systems *Full-Time Offer*

- Built GTFS-Realtime (Protobuf/HTTP) and TAIP (UDP) adapters in C#/.NET 10, feeding multi-agency bus telemetry into Hitachi's CAD dispatch; validated on MBTA's live feed ahead of a planned LA Metro rollout.
- Extended C++ gateway's TCP protocol to v3 to relay agency names to dispatch; fixed byte-alignment bugs.
- Architected and led a 13-intern depot bus tracker (YOLOv8-OBB, 2-stage OCR, homography) at 30 FPS on CPU with sub-second detection; verified all 15 fleet IDs and sent 4,009 updates in 4 min to SmartYard.
- Fixed a production LockRecursionException in SmartYard's C# by narrowing ReaderWriterLockSlim scope.
- Merged 10 PRs (65 commits, 40+ story points) through senior review; rated at Associate Engineer level.

Clever Devices (acquired by Hitachi Rail) June 2025 – Aug 2025

Software Engineer Intern *Return Offer*

- Built a multithreaded Python fuzzer firing synthetic POST payloads at a fault-tolerant vehicle REST API, uncovering unhandled exceptions; containerized the tool with Docker and charted each fuzzing run in Splunk.
- Built a real-time C# tracker service on the transit server that detects buses entering and leaving their depots.

Cloud Data Pro June 2024 – Aug 2024

Software Engineer Intern

- Ended a production outage by writing and deploying a SQL hotfix for the broken stored procedure behind it.
- Built a CI/CD pipeline on Azure deploying Docker images from Artifactory to Kubernetes with JWT auth.

Projects

Autonomous Micromouse Robot | *Team Captain* | *C++, Raspberry Pi Pico (RP2040), PID, CMake*

- Captained 13-member team for 2 seasons: 2nd of 15+ at MIT, 1st in Best Design, 2nd at NYIT (38s run).
- Built a double-buffered sensor hub across Pico's 2 cores (lock-free reads, mutex on publish) for 100 Hz PID.
- Built flood fill and A* with 45-degree diagonals and trapezoidal motion profiles; wrote 110 of 121 commits.
- Compiled the robot's C++ navigation code into a host simulator to rehearse all 3 competition run phases.

WorkIt | *5-Person Team* | *Python, FastAPI, Next.js, Supabase Postgres, GitHub Actions, Gemini*

- Built a robots.txt-compliant Python scraper reading 1,800 company boards (Greenhouse, Lever, Ashby) into 1,000+ roles; cut runtime from 9.5 to 3 min with one lane per provider and quiet boards read every 3rd run.
- Made the 6-hourly import one Postgres transaction that refuses an empty feed or closing over half the jobs.
- Shipped Scout, a FastAPI + Gemini assistant grounded only in a user's parsed resume and the job posting.

Skills

Languages: C++, C#, Python, TypeScript, JavaScript, Java, C, SQL
Frameworks: .NET, ASP.NET Core, FastAPI, Next.js, React, PyTorch, OpenCV, YOLO
Infrastructure: Docker, Kubernetes, Azure, AWS, RabbitMQ, PostgreSQL, Linux, Git, GitHub Actions
AI Tools: Claude Code, GitHub Copilot